Engineering Architecture Documentation 2024

Page 1 — Engineering Overview

The engineering department is responsible for designing, developing, and maintaining the company’s enterprise AI systems and cloud infrastructure. The platform architecture focuses on scalability, security, observability, and efficient retrieval pipelines.

The engineering team uses modern AI engineering practices to support enterprise-grade applications.

Page 2 — Backend Infrastructure

The backend system is built using:

- FastAPI
- Python
- LangChain

FastAPI provides scalable API routing and efficient asynchronous request handling for AI workflows.

Page 3 — Vector Database Systems

Semantic embeddings are stored in ChromaDB.

The vector database enables semantic similarity retrieval for enterprise RAG pipelines. Embedding models convert text chunks into vector representations for efficient retrieval.

Page 4 — Hybrid Retrieval Pipeline

The retrieval system combines:

- BM25 keyword search
- vector similarity search

This hybrid retrieval strategy improves search relevance and balances exact keyword matching with semantic understanding.

Page 5 — Reranking Systems

Retrieved chunks are passed through a reranking stage before prompt generation.

Reranking improves contextual relevance and helps reduce noisy retrieval results before sending context to the language model.

Page 6 — Security Architecture

The platform includes multiple enterprise security layers:

- Role Based Access Control (RBAC)
- metadata filtering
- query guardrails
- secure API routing

These systems help prevent unauthorized access to sensitive enterprise information.

Page 7 — Metadata Filtering

Document chunks contain metadata such as:

- department
- source file
- access level
- document type

Metadata-aware retrieval allows the system to filter documents before vector search execution.

Page 8 — API Deployment

The engineering team exposes enterprise RAG endpoints through FastAPI APIs.

Swagger documentation is automatically generated for API testing and integration workflows.

Page 9 — Monitoring and Observability

The platform integrates:

- LangSmith tracing
- RAGAS evaluation
- logging systems

These tools help monitor retrieval quality, response accuracy, and overall AI system performance.

Page 10 — Future Infrastructure Plans

Future engineering goals include:

- Docker deployment optimization
- Kubernetes orchestration
- streaming LLM responses
- agentic workflows
- LangGraph integration

The engineering department continues improving scalability, enterprise security, and AI workflow orchestration.